

Appendix B

EthOS Approval

EthOS Reference Number: 92134

Introduction

Thank you for visiting the Ethics Online System (EthOS)

All members of staff and students who are about to undertake a research project or investigation are required to use EthOS to determine what form of ethical approval, if any, is required.

The questions within EthOS will be automatically generated based on your previous answers so you may not see all sections.

Once you have submitted your application then it will receive a proportionate level of review based on your answers.

If, during review, any changes are required then the application will be returned to you. Once you've made the changes then you will be able to resubmit.

Please be aware that 'submission' does **not** constitute approval - you will need to wait until an approval letter has been issued (via email) by the system.

Applicant Information

Title

First Name

Surname

Email

A2 Manchester Metropolitan University ID number

Applicant Status

B1 Are you completing this application as a...

UG/PGT Student - Default application

B2 Please select the faculty that is most closely aligned with your project

Science & Engineering

B6 Please select the Science and Engineering department that is responsible for this project

School of Computing and Mathematics

B3.1 If you have selected 'Other' as your department, please specify

B1.2 Manchester Metropolitan Supervisor Details

First Name

Nishanthi

Surname

Abeynayake

Email

N.Abeynayake@mmu.ac.uk

Please click "Next" or "Navigate" on the sidebar to proceed with the application.

Project Details

D0.0 Please enter the module code for your course. This can be found on Moodle. If you are unsure, please contact your supervisor.

6G7V0007

D1 Full Project Title

Navigating the UK Property Market Predictive Analysis of House Price Index Trends with Affordability Modelling and Explainable Deep Learning

D2 Project Short Title

This should be the title shown in your Participant Information Sheet (PIS), consent form, advertising materials and any other supporting documents.

Navigating the UK Property Market Predictive Analysis of House Price Index Trends with Affordability Modelling and Explainable Deep Learning

D3 Estimated Start Date

20/05/2026

The start of your data collection must be in the future. It is not the date when you registered/started at the university. When selecting this date please allow time for your application to be reviewed.

D4 Project End Date

This should be the date when the final report for the study is submitted. If you are unsure of this date, please put the date when all aspects of the project will be complete.

For students, this is the final submission date of your thesis/dissertation/assignment.

11/09/2026

Please click "Next" or "Navigate" on the sidebar to proceed with the application.

UG/PGT Regulatory Screener

Begin UGPGT screening section

This section will ask a series of questions that will determine if the 'UG/PGT - Default Application' is the correct pathway to approval for you. Please answer these questions carefully.

If you are unsure how to answer the following questions then please contact your supervisor in the first instance. If you and your supervisor are having difficulty then please contact your [Faculty Ethics](#) team.

☒ Begin

UGPGT0 Does your project contain any of the following:

- Approval from an external regulatory body (including, but not limited to: NHS (HRA), HMPPS, MOD etc.);
- The use of ionising radiation or exposure to radioactive materials;
- Misleading participants;
- Research without the recruited participants' consent;
- The use of novel techniques, even where apparently non-invasive, whose safety may be open to question;
- Engaging in, witnessing, or monitoring criminal activity;
- Engaging with, or accessing terrorism related materials;
- A requirement for security clearance or licence to access participants, data or materials; Animals,
- animal tissue, new or existing human tissue, or biological toxins and agents;

☐ Yes

☐ Yes but I have discussed this question with my supervisor and have been advised to continue

☒ No

UGPGT1

Does your project contain any of the following:

- Processing of data relating to criminal convictions and offences
- Large Scale processing of sensitive special category data or data of a highly personal nature
- Processing of data concerning vulnerable individuals
- Use of MRI images or MRI data
- Systematic monitoring of a publicly accessible place on a large scale
- Processing involving the use of innovative technologies, or the novel application of existing technologies (such as Artificial Intelligence, Internet of Things (IoT), or location tracking)
- Processing of biometric or genetic data
- Matching or combining data sets from more than one source
- The processing of personal data which could result in a risk of harm to the data subject in the event of a security breach
- The processing of personal data without providing a privacy notice or participant information sheet directly to the individual

☐ Yes

☐

☒ Yes but I have discussed this question with my supervisor and have been advised to continue

No

UGPGT2

Does your project contain any of the following:

- Travel to countries considered "High Risk" on the Government's [Foreign Travel Advice \(FCDO\) website](#); Working with Hepatitis, Human T-Cell Lymphotropic Virus Type iii (HTLV iii), or Lymphadenopathy Associated Virus (LAV) or the mutants, derivatives or variations thereof, or Acquired Immune Deficiency Syndrome (AIDS), or any
- syndrome or condition of a similar kind; Working with Transmissible Spongiform Encephalopathy (TSE), Creutzfeldt-Jakob Disease (CJD), variant Creutzfeldt-
- Jakob Disease (vCJD) or new variant Creutzfeldt-Jakob Disease (nvCJD);
- Working in hazardous areas or high risk countries;
- Working with hazardous substances outside of a controlled environment;
- Working with persons with a history of violence, substance abuse or a criminal record.

☐

☐ Yes

☒ Yes but I have discussed this question with my supervisor and have been advised to continue

No

UGPGT3

Does your project contain any of the following:

- Physically invasive techniques;
- Ingestion of food, liquids, or drugs;
- Physical testing;
- Psychological testing;

in combination with any of the following:

- Pregnant women as participants;
- Children aged five and under; Blood
- borne pathogens;
- Cell Gene Therapy, therapies targeting blood, blood barrier or cerebrospinal fluid;
- Any Product, Pharmaceutica or Medical Device that does not have the appropriate regulatory approval;
- Activities being undertaken by the lead investigator or any other member of the study team in a country outside of the UK;

☐

☐ Yes

☒ Yes but I have discussed this question with my supervisor and have been advised to continue

No

Does your application contain any of the following:

- The use of equipment and/or substances that have not been used before, do not follow standard procedures and will be carried out in locations where a suitable and sufficient risk assessment is not already in place;
- Any hazardous activities, travel, substances, equipment, and/or locations that potentially pose a risk of ill health, injury or other harm (including stress) to participants, researchers or others;
- Any individuals or groups of people specifically at risk e.g. young persons, new and expectant mothers etc.;
- The need to obtain expert advice to ensure you have identified all reasonably foreseeable risks.

- ☐ Yes
- ☒ No

UG/PGT Application Information

Begin UGPGT Section

The following questions relate to the ethical concerns associated with your project.

- ☒ Begin section

UGPGT5 Is your project connected to another EthOS approved project?

- ☐ Yes
- ☒ No

UGPGT6 Please describe your research

Please make sure to discuss your aim, the activities you undertake, any special software/equipment you will use and the outcomes you expect.

This project looks at whether we can better predict housing affordability across UK regions not just raw house prices, but whether ordinary workers can actually afford to buy where they live. I do this by building a multi-task LSTM deep learning model that forecasts both the regional House Price Index and an affordability ratio (house prices relative to local median wages), then compare it against more traditional approaches like ARIMA and SARIMA to see what genuinely works best.

All the work is desk-based and computational. I pull together four freely available government datasets Land Registry house price data, ONS earnings figures, ONS macroeconomic indicators, and the Bank of England base rate clean and align them, then train and test the models. No people are involved at any point, and none of the data contains anything personal or identifiable.

Everything runs on open-source Python tools (pandas, statsmodels, tensorflow/keras, and shap) on a standard laptop, with Google Colab's free GPU where needed for the heavier model training. No specialist equipment is involved.

The end goal is a fully reproducible pipeline published on GitHub, a clear and honest comparison of how the different models perform across UK regions, and through SHAP analysis a readable explanation of which economic factors, like interest rates or wage growth, are actually driving affordability changes in different parts of the country.

UGPGT Human Participants

UGPGT7 Does your research involve recruiting human participants?

This includes social media content (with or without the account holder's knowledge)

- ☐ Yes
- ☒ No

UGPGT Disclosures and Conditions

UGPGT10 Please summarise the main ethical issues arising from your project and say how they have been addressed

The main ethical consideration for this project is data privacy specifically, whether the data I am using could raise any concerns about how people's information is handled. Honestly, this is not a concern here. Every dataset I use comes from UK government sources (Land Registry, ONS, and the Bank of England) and contains only aggregated statistics regional averages and national indicators. There are no individual records, no names, no addresses, and nothing that could ever be traced back to a real person. I am essentially working with spreadsheets of national economic figures, not anything personal.

Because the project is entirely computational and uses no human participants, there are no issues around consent, recruitment, deception, or psychological harm. Nobody is interviewed, surveyed, observed, or affected by this research in any direct way.

The one area worth flagging is that the project uses artificial intelligence (a deep learning model called an LSTM). This has been considered carefully. The AI is applied only to aggregate, non-personal economic data it is not making decisions about individuals or being used in any way that could harm or disadvantage anyone. It is a forecasting tool applied to publicly available statistics, which sits well within normal academic data science practice.

All data is handled and stored securely on my personal device and Google Colab, accessed only by me, and will not be shared beyond what appears in the dissertation and GitHub repository. The GitHub repository contains only code and aggregated outputs no raw personal data, because none exists in this project.

In short, the ethical risk profile of this project is minimal, and no special measures beyond standard good practice are required.

UGPGT11 By submitting this application you agree that you will do the following:

- If any of the details that you have declared in this application change then you will apply for an amendment;
- You will store all your project data (for which Manchester Met is Data Controller) in your student OneDrive account at Manchester Met;
- If your project uses personal data, you will store identifiable data separately from non-identifiable data and link them through a PIN (Participant Identification Number);
- You will destroy identifiable data according to the [University's retention and disposal schedule](#);
- You will not share identifiable data with any third-party without previously discussing this with dataprotection@mmu.ac.uk;
- You will maintain participant anonymity in publication unless the participant consented (in writing) to their identity being made public.

☒ I agree

Declarations

Y1 Do you have anything else you wish to declare or any documents you have not yet uploaded?

- ☒ Yes
- ☐ No

Y1.1 Please describe

Terms Of Reference

Y1.2 Please upload any additional documentation

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Additional Documentation	JERUSHAJOHANNAJANETRINJULIESDAVIDD ASS25867550	JERUSHAJOHANNAJANETRINJULIESDAVIDDASS 25867550.PDF	27/05/2026	1	411.1 KB

Signatures

Please be aware that the system will not process this application until all signatures (applicant/PI/supervisor) have been provided. Once all signatures have been provided then the application will be able to move to a 'submitted' status.

Please also be aware that 'submission' does not constitute approval - you will need to wait until an approval letter has been issued (via email) by the system. This email will arrive from 'donotreply@infonetica.net' - some email accounts will automatically sort this into your junk folder. This letter can also be downloaded in the system.

I confirm that all information in this application is accurate and true. I will not start this project until I have received an approval letter from the system.

☒ I confirm

Z2 Have you been instructed by your supervisor to request a second signature for this application?

☐ Yes

☒ No

Z3 Please click [here](#) to request your supervisor's signature. You will be asked to input their email address.

Your application will not be submitted until your supervisor has signed - it is your responsibility to ensure that they do this.

Signed: This form was signed by Nishanthi Abeynayake (N.Abeynayake@mmu.ac.uk) on 04/06/2026 08:46

Z6 By signing this application you are confirming that all details included in the form have been completed accurately and truthfully.

Signed: This form was signed by Jerusha Julies David Dass (JERUSHA.J.JULIES-DAVID-DASS@stu.mmu.ac.uk) on 03/06/2026 22:31